Example 8.27

A can solve 90 per cent of the problems given in a book and *B* can solve 70 per cent. What is the probability that at least one of them will solve a problem selected at random?

Solution:

Given the probability that A will be able to solve the problem $= \frac{90}{100} = \frac{9}{10}$ and the probability that B will be able to solve the problem $= \frac{70}{100} = \frac{7}{10}$

i.e., $P(A) = \frac{9}{10}$ and $P(B) = \frac{7}{10}$

$P(\bar{A}) = 1 - P(A) = \frac{9}{10} = \frac{1}{10}$

$P(\bar{B}) = 1 - P(B) = \frac{7}{10} = \frac{3}{10}$

P(at least one solve the problem) $= P(A \cup B)$

$= 1 - P(\overline{A \cup B}) = 1 - P(\bar{A} \cap \bar{B})$

$= 1 - P(\bar{A}) \cdot P(\bar{B})$

$= 1 - \frac{3}{100} = \frac{97}{100}$

Hence the probability that at least one of them will solve the problem $= \frac{97}{100}$.

Example 8.28

A bag contains 5 white and 3 black balls. Two balls are drawn at random one after the other without replacement. Find the probability that both balls drawn are black.

Solution:

Let *A*, *B* be the events of getting a black ball in the first and second draw.

Probability of drawing a black ball in the first attempt is

$P(A) = \frac{3}{5+3} = \frac{3}{8}$

Probability of drawing the second black ball given that the first ball drawn is black

$P(B/A) = \frac{2}{5+2} = \frac{2}{7}$

$\therefore$ The probability that both balls drawn are black is given by

$P(A \cap B) = P(A)\ P(B/A) = \frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$

Example 8.29

In a shooting test the probability of hitting the target are $\frac{3}{4}$ for *A*, $\frac{1}{2}$ for *B* and $\frac{2}{3}$ for *C*. If all of them fire at the same target, calculate the probabilities that

(i) All the three hit the target

(ii) Only one of them hits the target

(iii) At least one of them hits the target

Solution:

Given $P(A) = \frac{3}{4}$, $P(B) = \frac{1}{2}$, $P(C) = \frac{2}{3}$

Then $P(\bar{A}) = 1 - \frac{3}{4} = \frac{1}{4}$; $P(\bar{B}) = 1 - \frac{1}{2} = \frac{1}{2}$ and $P(\bar{C}) = 1 - \frac{2}{3} = \frac{1}{3}$

(i) $P(\bar{A})$=(all the three hit the targets)

$= P(A \cap B \cap C) = P(A)P(B)P(C)$

(since A, B, C hits independently)

$= \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{4}$

(ii) P (only one of them hits the target)

$= P\{(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)\}$

$= P\{(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C)\}$

$= \left(\frac{3}{4} \cdot \frac{1}{2} \cdot \frac{1}{3}\right) + \left(\frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{3}\right) + \left(\frac{1}{4} \cdot \frac{1}{2} \cdot \frac{2}{3}\right) = \frac{1}{4}$

(iii) P (at least one of them hit the target)

$= 1 -$ P(none of them hit the target)

$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$

$= 1 - P(\bar{A})P(\bar{B})P(\bar{C})$

$= 1 - \frac{1}{24} = \frac{23}{24}$

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